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Multiphysical modeling of transport phenomena during laser welding of dissimilar steels

A. Métais[a,b,*], S. Matteï[a], I. Tomashchuk[a], S. Gaied[b]

[a]Laboratoire Interdisciplinaire Carnot de Bourgogne (ICB), UMR 6303 CNRS / Université de Bourgogne Franche-Comté, 12 rue de la fonderie, 71200 Le Creusot, France

[b]ArcelorMittal, 1 route de Saint Leu, 60761 Montataire, France

Abstract

The success of new high-strength steels allows attaining equivalent performances with lower thicknesses and significant weight reduction. The welding of new couples of steel grades requires development and control of joining processes. Thanks to high precision and good flexibility, laser welding became one of the most used processes for joining of dissimilar welded blanks. The prediction of the local chemical composition in the weld formed between dissimilar steels in function of the welding parameters is essential because the dilution rate and the distribution of alloying elements in the melted zone determines the final tensile strength of the weld. The goal of the present study is to create and to validate a multiphysical numerical model studying the mixing of dissimilar steels in laser weld pool. A 3D modelling of heat transfer, turbulent flow and transport of species provides a better understanding of diffusion and convective mixing in laser weld pool. The present model allows predicting the weld geometry and element distribution. The model has been developed based on steady keyhole approximation and solved in quasi-stationary form in order to reduce the computation time. Turbulent flow formulation was applied to calculate velocity field. Fick law for diluted species was used to simulate the transport of alloying elements in the weld pool. To validate the model, a number of experiments have been performed: tests using pure 100 µm thick Ni foils like tracer and weld between a rich and poor manganese steels. SEM-EDX analysis of chemical composition has been carried out to obtain quantitative mapping of Ni and Mn distributions in the melted zone. The results of simulations have been found in good agreement with experimental data.

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Keywords: Laser welding; dissimilar materials; laminar flow; turbulent flow; transport of species

* Corresponding author. Tel.: +3-338-573-1017 .
E-mail address: alexandre.metais@arcelormittal.com

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1. Introduction

The proportion of Laser Welded Blanks (LWB) in vehicle design has significantly increased this last decade. Car manufacturers aim to develop more and more lightweight and strong car architectures that meet latest crash and safety requirements. The main benefits for LWB are mass and cost savings. Mass savings are achieved due to the higher strength level combinations. In order to achieve high-quality joining, it is important to have a deep understanding of the mixing process during full-penetrated laser welding. Nonetheless, many welding tests are often performed to identify operating parameters leading to a good joint. The use of numerical modeling allows in the same time to clarify the mechanism of weld formation and to attain important savings of cost and time during the determination of optimal operational parameters. The numerical capacity of current computers makes highly detailed simulation of laser welding possible. Over the last years, the complexity and accuracy of simulation models for laser processing have gradually increased from thermal model only (Kazemi and Goldak (2009)) to multiphysical model studying dynamic behavior of the keyhole (Courtois et al. (2013)).

The present paper is focused on mass transport mechanisms in the weld pool. Most of models assumed a laminar fluid flow in melted zone (Mahrle and Schmidt (2002); Tomashchuk et al. (2013); Courtois (2014)). Nevertheless, some simulations work with turbulent flow model even if Reynolds number order is about 100 (Rai et al. (2008); Wang et al. (2007); Chakraborty and Chakraborty (2007)). The aim of the present model is to carry out the best mathematical fluid flow model to predict mixing of alloyed elements in weld pool. Laminar, k-ε and k-ω models have been tested. For a long time, the development of fluid flow models of welding was hindered by a difficulty to elucidate the welding process by direct observation of flow in the weld pool. Recently, direct observation of inner process dynamic has shown to be possible with the use of high-speed imaging combined with advanced sensoring and illumination techniques (Boley et al. (2013)). In order to demonstrate the quantitative predictive capabilities of the present model, *post-mortem* observation involving chemical tracers was implied. Then, the model has been applied to the case of heterogeneous welds between poor and rich Mn steels.

Nomenclature

Cp^*	heat capacity without phase change $[J.kg^{-1}.K^{-1}]$
Cp	heat capacity $[J.kg^{-1}.K^{-1}]$
h_g	global heat exchange coefficient $[W.m^{-2}.K^{-1}]$
ρ	density $[kg.m^{-3}]$
ρ_p	plume density $[kg.m^{-3}]$
β	thermo-density coefficient $[K^{-1}]$
λ	conductivity $[W.m^{-1}.K^{-1}]$
T_0	ambient temperature $[K]$
T_f	melting point $[K]$
T_{vap}	boiling point $[K]$
L_f	melting latent heat $[J.kg^{-1}]$
ΔT	temperature range of 100 $[K]$
u	velocity field $[m.s^{-1}]$
V	welding velocity $[m.s^{-1}]$
V_p	velocity of vapor plume $[m.s^{-1}]$
γ	surface tension $[N.m^{-1}]$
γ_0	surface tension of pure iron $[N.m^{-1}]$
$d\gamma/dT$	surface tension gradient $[N.m^{-1}.K^{-1}]$
D	diffusion coefficient $[m^2.s^{-1}]$
c	concentration $[mol.m^{-3}]$
ν_T	turbulent cinematic viscosity $[m^2.s^{-1}]$
k_B	Boltzmann constant $[J.K^{-1}]$
μ	dynamic viscosity $[Pa.s]$
r	radius of the particle $[m]$
g	gravity constant $[m.s^{-2}]$
p	pressure $[Pa]$
Re	Reynolds number
k	turbulent kinetic energy
ε	rate of dissipation of kinetic energy
ω	specific rate of dissipation of kinetic energy

2. Experimental part

Laser beam welding was carried out with a Yb:YAG laser, a power of 4 kW, a welding velocity of 6 m/min and a spot size of 600 μm. To compare different fluid flow models, three cases have been tested experimentally as well as numerically.

In two cases, welding of homogeneous steel with pure nickel foils (100 µm thick) introduced between the welded plates (Figure 1-(a)) was performed. In the first case, the laser beam is centered on the nickel foil. For the second case, a 200 µm offset distance of the laser beam from the middle of the foil is applied. In theses cases, our approach (as described by Dörfler (2008)) is to add a metal foil that can be used as a tracer. Nickel was chosen as tracer to reveal the mixing process in the melted zone because of its similarity with iron in term of density, thermal conductivity and heat capacity. It is supposed that the addition of a small quantity of nickel in the melted zone does not modify significantly the local thermophysical properties. 100 µm thick foils are sufficient to ensure good traceability with SEM-EDX analysis. In the third case a manganese rich steel (22 %ωt Mn) is joined with steel containing 1.5 %ωt Mn. The laser beam is centered with the joint line (Figure 1-(b)).

Post-mortem mixing map formed between dissimilar metals is used to conclude on the flow in the molten pool. Indeed, it is supposed that the transport of species provides a representative global image of the convection phenomena in the weld pool. Based on *post-mortem* X-mapping of addition element in weld cross-sections, several hypothesis on flow in the weld pool can be made. EDS element analysis was performed with a JEOL JSM-6610LA field emission SEM equipped with an JED-2300F EDX system.

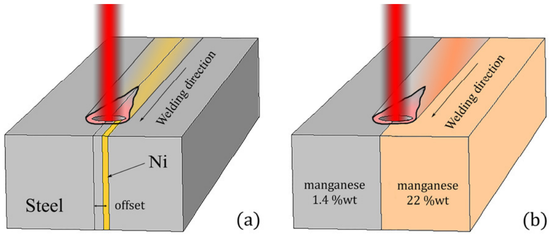

Fig. 1. Welding configuration cases 1, 2 (a) and case 3 (b).

3. Model description

3.1. General assumptions

A number of assumptions have been made to develop the following tridimensional model with the goal to reduce the computation time:

- a steady keyhole with a conical geometry;
- temperature inside the keyhole is assumed to be uniform $T_{keyhole} = T_{vap}$;
- top and bottom surfaces of the weld are assumed to be flat;
- heat transfer and fluid flow equations are strongly coupled;
- liquid metal is assumed to be Newtonian and incompressible;
- quasi-steady approach is used.

3.2. Geometry and mesh

During keyhole laser welding, the focused laser beam irradiates the top surface of the steel sheet, inducing the formation of melted zone. Because of the very high energy density of the laser, the material quickly vaporizes. The created vapor pushes the liquid to form a vapor capillary, named a keyhole. The interaction between recoil pressure and other driving forces controls the stability of the keyhole. A steady-state is assumed for the present study. The keyhole is represented by a cone-shaped geometry. The maximal diameter of the cone (top surface) is equal to the diameter of focused laser spot (600 μm) and the minimal diameter (bottom surface) is equal to 500 μm. The sheet thickness is conserved (1.6 mm). Two meshes are used for the calculation; a coarse mesh (Figure 2-(a)) to obtain quickly a “good” initial condition and a fine mesh (Figure 2-(b)) to solve the whole of equations using the previously found solution. A representative volume of the welded blanks is meshed. Tetrahedral mesh of 60 μm maximal size was additionally refined around the keyhole.

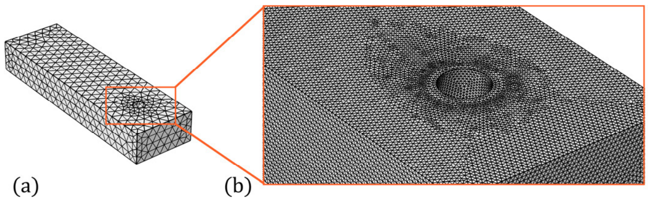

Fig. 2. Coarse mesh (a) and fine meh (b).

3.3. Material properties

One of the most important features of the dissimilar weld simulation is handling the material properties. The material properties are defined as function of location and temperature. Smoothed "Heaviside step functions" are employed to smooth sharp changes of material properties at the boundary of dissimilar metals and in the phase change zone (example on Figure 3). The values of material properties are presented in Table 1.

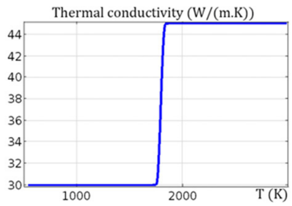

Fig. 3. Thermal conductivity of 1.4 ωt% Mn steel with Heaviside step function.

Table 1. Material properties.

Property	1.4 ωt% Mn steel [solid, liquid]	22 ωt% Mn steel [solid, liquid]
Fusion temperature (K)	1798	1648
Heat of fusion $(kJ.kg^{-1})$	270	270
Heat capacity $(J.kg^{-1}.K^{-1})$	[500, 600]	[550, 650]
Density $(kg.m^{-3})$	$7800/(1+\beta \cdot T)$	$7500/(1+\beta \cdot T)$
Thermo-density coefficient β (K^{-1})	$15 \cdot 10^{-6}$	$20 \cdot 10^{-6}$
Thermal conductivity $(W.m^{-1}.K^{-1})$	[30, 45]	[40, 50]
Viscosity $(Pa.s)$	[100, 0.003]	[100, 0.003]

4. Governing equations

Heat transfer, flow and transport of diluted species problems are solved with COMSOL Multyphysics® 5.1. In segregated resolution, direct PARDISO solvers are used to calculate variable T and iterative GMRES solver is used to calculate fluid flow and transport species variables. The use of iterative solver allows reducing the use of RAM.

4.1. Heat transfer

The energy equation solved with convection term provides temperature field over the entire domain. Because of the quasi-steady approach, the velocity field $\boldsymbol{u}$ has a constant term V in the welding direction added at the calculated velocity field.

$$\rho C_p \boldsymbol{u} \cdot \nabla T + \nabla(-\lambda \nabla T) = 0 \quad (1)$$

The fusion is taken into account by using equivalent enthalpy method

$$C_p(T) = C_p^*(T) + \delta \cdot L_f \tag{2}$$

where δ is a Gaussian distribution defined as:

$$\delta = \frac{1}{\Delta T \cdot \sqrt{\pi}} \cdot e^{\frac{T-T_f}{\Delta T}} \tag{3}$$

4.2. Fluid flow

Three different flow models have been tested in order to conclude on the best choice to simulate full penetrated laser welding of dissimilar steels. The governing equations for mass conservation, momentum and energy transport in steady-state formulation are as follows as they are implemented within the simulation framework of COMSOL Multiphysics® 5.1.

Because of the Reynolds number close to 100 for a flow around a cylinder, a flow regime of transition can exist. For Re << 1 the flow is laminar and Re > 1000 the flow is fully turbulent. Between these two values intermediate flow regimes can be observed like recirculating flow or vortexes making Karman's paths (Guyon et al. (1994)). The following Navier-Stokes equations were first of all tested in simulation:

$$\begin{gathered} \rho\nabla \cdot (\boldsymbol{u}) = 0 \\ \rho(\boldsymbol{u} \cdot \nabla)\boldsymbol{u} = \nabla \cdot [-p\boldsymbol{I} + \mu \cdot (\nabla\boldsymbol{u} + (\nabla\boldsymbol{u})^T)] - \rho\boldsymbol{g} + F^M \end{gathered} \tag{4}$$

Then, Reynolds-Averaged Navier-Stockes turbulent model have been used in order to take into account turbulent mixing caused by eddy diffusivity. On the one hand, the $k - \varepsilon$ model is very popular for industrial applications due to its good convergence rate and relatively low memory requirements. Nevertheless, it does not very accurately compute flow fields that exhibit adverse pressure gradients, strong curvature to the flow, or jet flow. It does perform well for external flow problems around complex geometries. On the other hand, the $k - \omega$ model is useful in many cases where the $k - \varepsilon$ model is not accurate, such as internal flows, flows that exhibit strong curvature, separated flows, and jets. Since top and bottom surfaces are assumed to be flat, the numerical model simulates an internal flow and thus $k - \omega$ model seems to be better. However, both turbulent models have been performed to conclude on the best model to use. The following equations are about the $k - \varepsilon$ model,

$$\begin{gathered} \rho\nabla \cdot (\boldsymbol{u}) = 0 \\ \rho(\boldsymbol{u} \cdot \nabla)\boldsymbol{u} = \nabla \cdot [-p\boldsymbol{I} + (\mu + \mu_T)(\nabla\boldsymbol{u} + (\nabla\boldsymbol{u})^T)] - \rho\boldsymbol{g} + F^M \\ \rho(\boldsymbol{u} \cdot \nabla)k = \nabla \cdot \left[\left(\mu + \frac{\mu_T}{\sigma_k}\right)\nabla k\right] + p_k - \rho\varepsilon \\ \rho(\boldsymbol{u} \cdot \nabla)\varepsilon = \nabla \cdot \left[\left(\mu + \frac{\mu_T}{\sigma_\varepsilon}\right)\nabla \varepsilon\right] + C_{\varepsilon 1}\frac{\varepsilon}{k}p_k - C_{\varepsilon 2}\rho\frac{\varepsilon^2}{k} \\ \mu_T = \rho C\mu\frac{k^2}{\varepsilon} \\ p_k = \mu_T[\nabla\boldsymbol{u} : (\nabla\boldsymbol{u} + (\nabla\boldsymbol{u})^T\ )\ ] \end{gathered} \tag{5}$$

and the last equations about the $k - \omega$ model.

$$
\begin{gathered}
\rho \nabla \cdot (\boldsymbol{u}) = 0 \\
\rho(\boldsymbol{u} \cdot \nabla)\boldsymbol{u} = \nabla \cdot [-p\boldsymbol{I} + (\mu + \mu_T)(\nabla \boldsymbol{u} + (\nabla \boldsymbol{u})^T)] - \rho \boldsymbol{g} + F^M \\
\rho(\boldsymbol{u} \cdot \nabla)k = \nabla \cdot [(\mu + \mu_T \sigma_k^*)\nabla k] + p_k - \beta_0^* \rho \omega k \\
\rho(\boldsymbol{u} \cdot \nabla)\omega = \nabla \cdot [(\mu + \mu_T \sigma_\omega)\nabla \omega] + \alpha \frac{\omega}{k} p_k - \rho \beta_0 \omega^2 \\
\mu_T = \rho \frac{k}{\omega} \\
p_k = \mu_T [\nabla \boldsymbol{u} : (\nabla \boldsymbol{u} + (\nabla \boldsymbol{u})^T)]
\end{gathered} \tag{6}
$$

The moving solid-liquid interface is modeled through changing the viscosity of the fluid. Dynamic viscosity is temperature dependent: solid is assumed as an equivalent liquid with a viscosity of 100 $Pa.s$.

4.3. Fick Law for diluted species

To study transport of species in weld pool, Fick law has been used,

$$\nabla \cdot (-D_i \nabla c_i + \boldsymbol{u} c_i) = 0 \tag{7}$$

where D_i the diffusion coefficient of the element i is defined by the following equation:

$$D_i(T) = D_{thermal}(T) + D_{turbulent}(T) = \frac{k_B T}{6\pi r_i \mu} + \frac{\sqrt{2}}{2} \nu_T \tag{8}$$

The term $D_{turbulent}(T)$ equals zero when the laminar flow condition is solved. Density, conductivity and heat capacity are considered as independent of species.

4.4. Boundary conditions

The input power is modeled by a uniform temperature on the keyhole walls. Symmetric condition is applied on lateral walls. The heat is exchanged between the weld plates and environment, consequently heat losses due to convection and radiation are taken into account throughout a global heat exchange coefficient on top and bottom surfaces.

$$\lambda \vec{\nabla} T \cdot \vec{n} = -h_g (T - T_0) \tag{9}$$

Inflow condition at welding velocity is imposed on front surface and outflow condition at constant pressure on back surface. Sliding walls are imposed on whole surfaces. Due to the variation of the surface tension coefficient with temperature,

$$\gamma = \gamma_0 + \frac{d\gamma}{dT} \cdot (T - T_f) \tag{10}$$

Marangoni effect, which is one of driving forces in the weld pool, is taken into account on top and bottom surfaces.

$$F^M = \frac{\partial \gamma}{\partial T} \cdot \frac{\partial T}{\partial \tau} \tag{11}$$

Full-penetrated welding produces a high velocity jet of gas out of the keyhole. In the literature, a velocity of the metallic vapour gas, named plume, higher than 100 $m.s^{-1}$ can be found (Amara and Bendid (2002)). The shear stress induced by the interaction of the plume on the liquid metal can't be neglected and has been introduced to surfaces of the keyhole as a weak contribution. This phenomenon is still misunderstood and in this model the shear stress is assumed to be a constant. Considering that the flow inside the vapor plume is laminar, the Darcy-Weisback equation gives an order of magnitude for the shear stress between liquid metal and vapor plume:

$$\tau_p = \frac{1}{8} f \rho_p V_p^2 \tag{12}$$

where $f = 64/Re$ for a laminar flow. A velocity of the plume $V_p = 100\ m.s^{-1}$ gives a shear stress of 50 $N.m^{-2}$. This value is currently used in our numerical models without experimental validation.

The concentration c from the Fick law is limited at the inflow surface. Others surfaces, except outflow surface, are imposed to no-flux condition.

5. Results and discussion

To validate our approach, the numerical results have been confronted to experimental tests.

5.1. Weld shape

An hourglass weld shape is generally observed for full-penetrated laser welds. Numerically, a negative temperature coefficient of surface tension results in hourglass weld shape. The metal flows from the higher-temperature keyhole boundary to the lower-temperature molten pool boundary and leads to the expansion of the fusion zone at the top and the bottom surfaces (Figure 4). Taking into account the hourglass shape of weld cross-section, three characteristic dimensions of melted zone were chosen to control the concordance of calculated and experimental solidus line: weld width is measured on top (L1) and bottom (L2) surfaces and in the middle (L3) of melted zone.

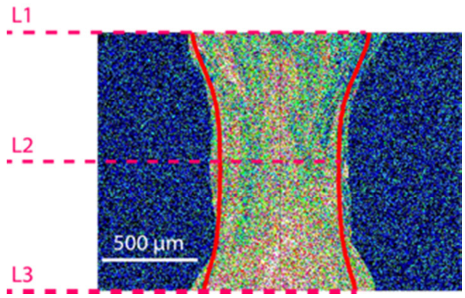

Fig. 4. Experimental vs numerical melted zones for case 2.

Table 2. 2D cross sections (μm) of experimental and calculated melted zones comparable.

Dimensions (μm)		Case 1			Case 2			Case 3		
		lam	$k-\varepsilon$	$k-\omega$	*lam*	$k-\varepsilon$	$k-\omega$	*lam*	$k-\varepsilon$	$k-\omega$
L1	Exp	858	858	858	1009	1009	1009	722	722	722
	Calc	822	802	768	822	802	768	841	818	789
	$\boldsymbol{\delta}$(%)	**4.2**	**6.5**	**10.5**	**18.5**	**20.5**	**23.8**	**-16.5**	**-13.3**	**-9.3**
L2	Exp	659	659	659	758	758	758	670	670	670
	Calc	708	717	720	708	717	720	722	727	728
	$\boldsymbol{\delta}$(%)	**-7.4**	**-8.8**	**-9.3**	**6.6**	**5.4**	**5.0**	**-7.8**	**-8.5**	**-8.6**
L3	Exp	1033	1033	1033	945	945	945	920	920	920
	Calc	846	855	839	846	855	839	831	830	810
	$\boldsymbol{\delta}$(%)	**18.1**	**17.2**	**18.8**	**10.5**	**9.5**	**11.2**	**9.7**	**9.8**	**11.9**

First, because of the hypothesis that material properties are independent of species concentration, numerical results on the weld dimensions have been found equal for the case 1 and 2. Then, the different flow models give similar results. Relative error inferior to 25% is observed (Table 2). Thus, the study of weld shape dimensions does not discriminate one of the three flow models tested.

5.2. Flow field in the weld pool

Numerical simulations illustrate that four eddies are formed near the top and bottom surfaces due to Marangoni convection (Figure 5). It can be noticed that the velocity in the weld pool is higher in these eddies, in particular next to top and bottom surfaces. Thermocapillary effects accelerate the fluid up to $0.5\ m.s^{-1}$ (ie. five time the laser welding velocity). Because of a negative temperature coefficient of surface tension, the flow goes from the center to the boundaries of the weld pool. In cross-section (Figure 5), the difference between the three flow models is small, the velocity magnitude is higher with laminar flow but main flows are similar.

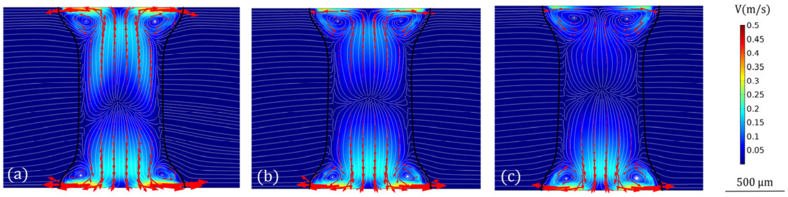

Fig. 5. Stream lines in a cross-section having maximal melted zone for case 3, color scale is the velocity magnitude $(\boldsymbol{m.s^{-1}})$, in black the melting point and red arrow the velocity field. Laminar flow (a), turbulent $\boldsymbol{k-\varepsilon}$ flow (b) and turbulent $\boldsymbol{k-\omega}$ flow (c).

In longitudinal observation, two eddies can be observed in y-z plane (Figure 6). The plume, a high velocity jet of gas, generates a strong shear stress and sets the liquid in motion. These eddies seem to be main driving force for mixing in z direction. The vortex size and the velocity field inside the vortex depend on this plume. Once again the velocity magnitude is higher with laminar flow, $1\ m.s^{-1}$ has been found on surface of the weld pool. Nevertheless, in longitudinal section the laminar model (Figure 6 – (a)) shows one more vortex than the turbulent models. In the following section, the importance of convective term in Fick Law low will be discussed. Thus, it has to be noted that the laminar flow solves big vortex with high degree of tolerance but does not solve eddies which are smaller than the mesh size; and the turbulent models solve an average flow field in melted pool.

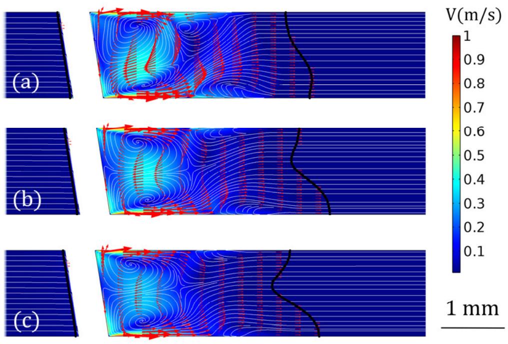

Fig. 6. Stream lines in a longitudinal section for case 3, color scale is the velocity magnitude $(\boldsymbol{m.s^{-1}})$ and red arrow the velocity field. Laminar flow (a), turbulent $\boldsymbol{k-\varepsilon}$ flow (b) and turbulent $\boldsymbol{k-\omega}$ flow (c).

5.3. Transport of diluted species

Numerical results show that thermal diffusive transport can be neglected in this study because of the high cooling rate. A relative error below 1% between calculations with thermal diffusion coefficient of $10^{-8}\ m^2.s^{-1}$ and $10^{-20}\ m^2.s^{-1}$ has been found. It is an essential result for this study because the convective term and the turbulent diffusive term in the Fick law are preponderant compared to the thermal diffusive term. The convective term is directly proportional to the velocity field. The convection problem has to be solved with a high level of precision because the quality and realism of the resulting velocity field has a direct impact on the correspondence between numerical and experimental element mapping. The challenge is to validate numerical flow field with help of experimental observations. *Post-mortem* X-mapping of alloying element like nickel and manganese is used to conclude on matter transport mechanisms.

First, Fick law calculations do not converge for tests working with turbulent $k-\varepsilon$ flow model even if the same default solver between all nine numerical tests is used. The only difference in Fick law equation for the three tests calculated with $k-\varepsilon$ Navier-Stockes model is the diffusive term in equation 12. The cinematic viscosity ν_T (calculated with $k-\varepsilon$ model) seems to produce oscillation of Fick law solution. Thus the $k-\varepsilon$ model is not adapted to this study and only the laminar and $k-\omega$ models will be studied to find the best flow model for studying laser welding between dissimilar materials (Figure 7).

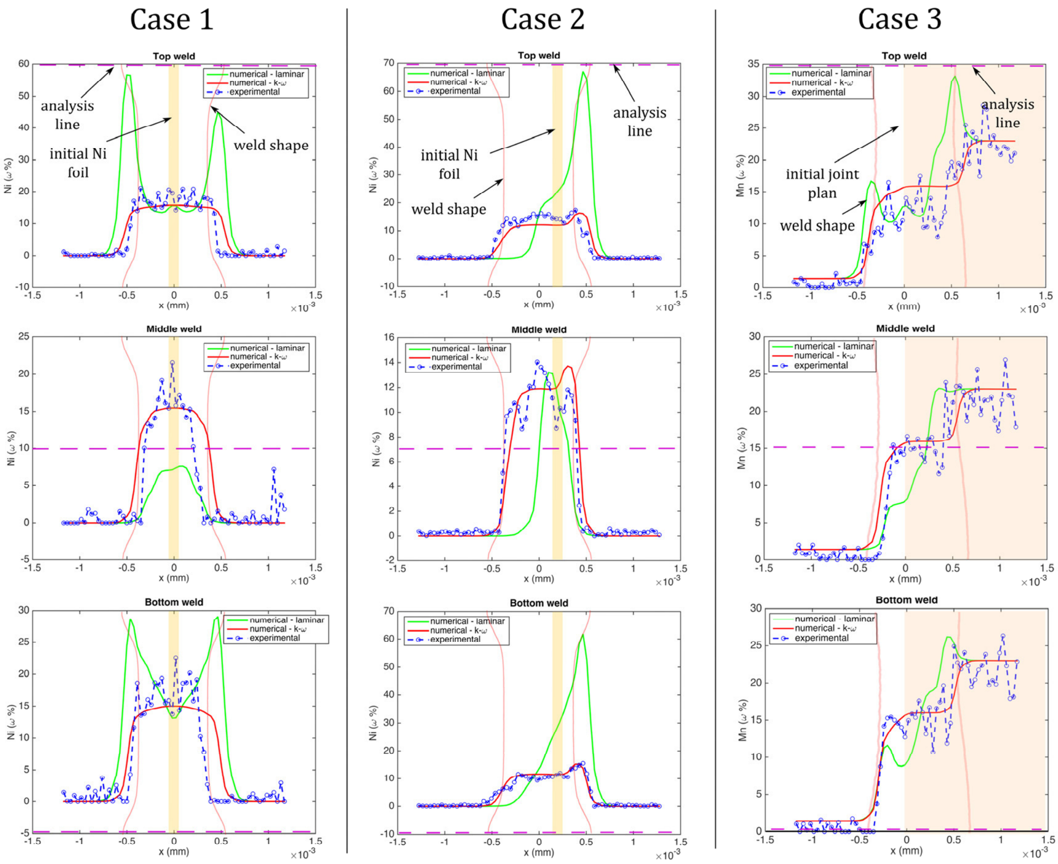

Fig. 7. Experimental and numerical distributions of alloying elements ($\omega t\%$) along the line L1, L2 and L3 for the three cases. Ni mass fraction for case 1 and 2, and Mn mass fraction for case 3.

Peaks of concentration can be observed near the top and bottom surfaces of the weld for numerical model solved with a laminar flow. These peaks coincide with localization of eddies formed by the Marangoni effect. Thus without use of turbulent diffusion, simulations show an increase of concentration of alloying elements in the center of biggest eddies. The calculation with $k-\omega$ model is essential to obtain an important mixing of alloying elements in the melted zone. Indeed, only a turbulent diffusivity gives a good mixing in the weld pool. Laminar flow does not calculate small eddies if the mesh size is not smaller than the size of these eddies. However, it is in these small eddies that the main part of mixing occurs. The dynamic viscosity from the $k-\omega$ model has to be calculated in order to access to the turbulent diffusion coefficient in the weld pool (Figure 8). This term (μ_T) becomes the predominant term of Fick law near the center of biggest eddies and at the interface between these biggest eddies.

Numerical results have been found in good agreement with experimental data for $k-\omega$ model (Figure 7). For all considered cases, good representation of alloying element distribution is observed. In case of Ni foil melting, for both positions of the foil, the dilution of Ni around 15 ωt% was attained, which is in good correspondence with results of EDX analysis. In case three, the weld formed between steels containing 22 and 1.5 ωt% Mn showed Mn dilution of 15 ωt% both numerically and experimentally. To obtain this result, the difference of melting points between these two steels has to be taken into account. Basing on Fe-Mn phase diagram Fe-Mn, the melting point of 22 ωt% Mn steel was considered to be 150 K inferior to the melting point of the 1.5 ωt% Mn steel.

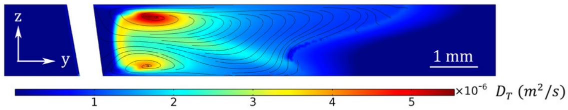

Fig. 8. Turbulent diffusion coefficient in the initial joint plane of the case 2 solved with $\boldsymbol{k - \omega}$ model.

6. Conclusion

A 3D model of full penetrated laser welding between dissimilar materials has been developed. Weld shape dimensions have been found in good agreement with experiment. Hourglass shape of the weld crosscut is observed due to Marangoni effect.

Turbulent $k - \omega$ model was found to be the most adequate for solving the convection problem compared to laminar and $k - \varepsilon$ model. Turbulent fluid convection is the main driving force of the mixing process. The biggest eddies formed by Marangoni convection and by shear stress of plume allow important displacement of particles in the pool. Four eddies form next to top and bottom surfaces due to Marangoni effect, and generate a flow going from the center to the boundaries of the weld pool with the highest velocity on surfaces. Plume shear stress contributes to two important eddies in longitudinal plane. Then small eddies, that are not calculated in this model but estimated by turbulent variables k and ω, allow to the particles to move from one major eddy to another.

Post-mortem X-mapping of alloying elements on weld cross section have been shown efficient to conclude on flow mechanisms in melted zone by reverse engineering. Numerical results obtain with $k - \omega$ model have been found in good agreement with experimental results in the three cases without changing any parameter except material properties.

This model is robust to the change of materials welded and preliminary results show a good behavior about change of welding parameters like welding speed, laser power or laser beam position from the initial line joint.

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